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CONDUIT SYSTEM, RADIATION SOURCE, LITHOGRAPHIC APPARATUS, AND METHODS THEREOF

Abstract

A pulsed-discharge radiation source includes a gas chamber, a window, and a conduit system. The conduit system includes a refill path and a conduit. The pulsed-discharge radiation source generates radiation. The gas chamber confines a gas and contaminants produced during the generation of radiation. The window isolates the gas from an environment external to the gas chamber and allows the radiation to travel between the gas chamber and the environment. The refill path allows a replacement of the gas. The conduit circulates the gas to or from the gas chamber during the generating. The conduit system directs a flow of one of a refill gas, the gas, or the refill gas and the gas at least during a refill operation to prevent the contaminant from contacting the window, whereby the conduit system increases the usable lifetime of at least the window.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/912,996, filed Sep. 20, 2022, which is the national phase of International Application No. PCT/US2021/021543, filed Mar. 9, 2021, which claims priority to U.S. Application No. 63/005,845 filed Apr. 6, 2020 and titled CONDUIT SYSTEM, RADIATION SOURCE, LITHOGRAPHIC APPARATUS, AND METHODS THEREOF, which is incorporated herein in its entirety by reference.

FIELD

[0002] The present disclosure relates to pulsed-discharge radiation sources, for example, an ultraviolet gas discharge laser for lithographic systems.

BACKGROUND

[0003] Methods to produce deep ultraviolet (DUV) radiation include, but are not limited to, using a pulsed-discharge radiation source. An excimer laser is an example of a pulsed-discharge radiation source. Pulsed-discharge radiation sources excite gas molecules confined in a chamber to generate laser radiation of a desired wavelength. The radiation can be let out of the chamber through a window. The gas molecules can include, but are not limited to, fluorine, neon, krypton, argon, and the like. The gas molecules may be excited by supplying a voltage (e.g., an electrical pulse) to the gas via electrodes. Over the course of the radiation source's lifetime, the gas chamber may develop contaminant particles due to the interaction of the electrodes and the gas. The contaminant particles may then contaminate other optically sensitive parts (e.g., the window) and cause unexpected, early failure of the radiation source.

[0004] Pulsed-discharge radiation sources may be utilized for generating radiation in a variety of applications, for example, generating DUV radiation in lithographic apparatuses. A lithographic apparatus is a machine that applies a desired pattern onto a substrate, usually onto a target portion of the substrate. A lithographic apparatus can be used, for example, in the manufacture of integrated circuits (ICs). In that instance, a patterning device, which may be a mask or a reticle, can be used to generate a circuit pattern to be formed on an individual layer of the IC. This pattern can be transferred onto a target portion (e.g., comprising part of, one, or several dies) on a substrate (e.g., a silicon wafer). Transfer of the pattern is typically via imaging onto a layer of radiation-sensitive material (resist) provided on the substrate. In general, a single substrate will contain a network of adjacent target portions that are successively patterned. Known lithographic apparatuses include so-called steppers, in which each target portion is irradiated by exposing an entire pattern onto the target portion at one time, and so-called scanners, in which each target portion is irradiated by scanning the pattern through a radiation beam in a given direction (the “scanning”-direction) while synchronously scanning the target portions in a direction parallel to and along the scanning direction, or parallel to and opposite the scanning direction.

[0005] A lithographic apparatus typically includes an illumination system that conditions radiation generated by a radiation source before the radiation is incident upon the patterning device. A patterned beam of DUV or light can be used to produce extremely small features on a substrate.

The illumination system may include a pulsed-discharge radiation source having a gas chamber that can be susceptible to early failure due to contaminants in the gas chamber.

SUMMARY

[0006] Accordingly, it is desirable to protect optically sensitive components or otherwise reduce the likelihood of untimely failure of a pulsed-discharge radiation source due to contaminants, for example, by managing the flow of gas within the radiation source.

[0007] In some embodiments, a pulsed-discharge radiation source comprises a gas chamber, a window, and a conduit system. The conduit system comprises a refill path and a conduit. The pulsed-discharge radiation system is configured to generate radiation. The gas chamber is configured to confine a gas and a contaminant produced during the generating. The window is configured to isolate the gas from an environment external to the gas chamber and to allow the radiation to travel between the gas chamber and the environment. The refill path is configured to allow a replacement of the gas. The conduit is configured to circulate the gas to or from the gas chamber during the generating. The conduit system is configured to direct a flow of one of a refill gas, the gas, or the refill gas and the gas at least during a refill operation to prevent the contaminant from contacting the window, whereby the conduit system increases the usable lifetime of at least the window.

[0008] In some embodiments, a method comprises generating radiation using a pulsed-discharge radiation source, confining a gas and contaminants produced during the generating using a gas chamber, isolating the gas from an environment external to the gas chamber using a window, allowing the radiation to travel between the gas chamber and the environment using the window, replacing the gas using a refill path, circulating the gas to or from the gas chamber during the generating, and directing a flow of one of a refill gas, the gas, or the refill gas and the gas during a refill operation to prevent the contaminant from contacting the window.

[0009] In some embodiments, a lithographic apparatus comprises an illumination system and a projection system. The illumination system comprises a gas chamber, a window, and a conduit system. The illumination system is configured to illuminate a pattern of a patterning device. The gas chamber is configured to confine a gas and a contaminant produced during the generating. The window is configured to isolate the gas from an environment external to the gas chamber and to allow the radiation to travel between the gas chamber and the environment. The refill path is configured to allow a replacement of the gas. The conduit is configured to circulate the gas to or from the gas chamber during the generating. The conduit system is configured to direct a flow of one of a refill gas, the gas, or the refill gas and the gas at least during a refill operation to prevent the contaminant from contacting the window, whereby the conduit system increases the usable lifetime of at least the window. The projection system is configured to project an image of the pattern onto a substrate.

[0010] Further features of the present disclosure, as well as the structure and operation of various embodiments, are described in detail below with reference to the accompanying drawings. It is noted that the present disclosure is not limited to the specific embodiments described herein. Such embodiments are presented herein for illustrative purposes only. Additional embodiments will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the relevant art(s) to make and use embodiments described herein.

[0012] FIG. 1A shows a reflective lithographic apparatus, according to some embodiments.

[0013] FIG. 1B shows a transmissive lithographic apparatus, according to some embodiments.

[0014] FIG. 2 shows a schematic of a lithographic cell, according to some embodiments.

[0015] FIGS. 3 and 4 show radiation sources, according to some embodiments.

[0016] FIGS. 5A, 5B and 5C each show a portion of a radiation source, according to some embodiments.

[0017] FIG. 6 shows a unidirectional valve, according to some embodiments.

[0018] FIG. 7 shows a cross-section of a unidirectional valve, according to some embodiments.

[0019] FIG. 8 is a flow chart that shows method steps for performing functions of embodiments described herein, according to some embodiments.

[0020] The features of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears. Unless otherwise indicated, the drawings provided throughout the disclosure should not be interpreted as to-scale drawings.

DETAILED DESCRIPTION

[0021] This specification discloses one or more embodiments that incorporate the features of the present disclosure. The disclosed embodiment(s) are provided as examples. The scope of the present disclosure is not limited to the disclosed embodiment(s). Claimed features are defined by the claims appended hereto.

[0022] The embodiment(s) described, and references in the specification to “one embodiment,” “an embodiment,” “an exemplary embodiment,” “an example embodiment,” etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic.

Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is understood that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0023] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “on,” “upper” and the like, can be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0024] The term “about” as used herein indicates the value of a given quantity that can vary based on a particular technology. Based on the particular technology, the term “about” can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., +10%, +20%, or +30% of the value).

[0025] Embodiments of the disclosure can be implemented in hardware, firmware, software, or any combination thereof. Embodiments of the disclosure may also be implemented as instructions stored on a machine-readable medium, which can be read and executed by one or more processors. A machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computing device). For example, a machine-readable medium may include read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; flash memory devices; electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.), and others. Further, firmware, software, routines, and/or instructions can be described herein as performing certain

actions. However, it should be appreciated that such descriptions are merely for convenience and that such actions in fact result from computing devices, processors, controllers, or other devices executing the firmware, software, routines, instructions, etc.

[0026] Before describing such embodiments in more detail, however, it is instructive to present an example environment in which embodiments of the present disclosure can be implemented.

Example Lithographic Systems

[0027] FIGS. **1A** and **1B** show schematic illustrations of a lithographic apparatus **100** and lithographic apparatus **100'**, respectively, in which embodiments of the present disclosure can be implemented. Lithographic apparatus **100** and lithographic apparatus **100'** each include the following: an illumination system (illuminator) IL configured to condition a radiation beam B (for example, deep ultra violet or extreme ultra violet radiation); a support structure (for example, a mask table) MT configured to support a patterning device (for example, a mask, a reticle, or a dynamic patterning device) MA and connected to a first positioner PM configured to accurately position the patterning device MA; and, a substrate table (for example, a wafer table) WT configured to hold a substrate (for example, a resist coated wafer) W and connected to a second positioner PW configured to accurately position the substrate W. Lithographic apparatus **100** and **100'** also have a projection system PS configured to project a pattern imparted to the radiation beam B by patterning device MA onto a target portion (for example, comprising one or more dies) C of the substrate W. In lithographic apparatus **100**, the patterning device MA and the projection system PS are reflective. In lithographic apparatus **100'**, the patterning device MA and the projection system PS are transmissive.

[0028] The illumination system IL can include various types of optical components, such as refractive, reflective, catadioptric, magnetic, electromagnetic, electrostatic, or other types of optical components, or any combination thereof, for directing, shaping, or controlling the radiation beam B. The support structure MT holds the patterning device MA in a manner that depends on the orientation of the patterning device MA with respect to a reference frame, the design of at least one of the lithographic apparatus **100** and **100'**, and other conditions, such as whether or not the patterning device MA is held in a vacuum environment. The support structure MT can use mechanical, vacuum, electrostatic, or other clamping techniques to hold the patterning device MA. The support structure MT can be a frame or a table, for example, which can be fixed or movable, as required. By using sensors, the support structure MT can ensure that the patterning device MA is at a desired position, for example, with respect to the projection system PS.

[0029] The term “patterning device” MA should be broadly interpreted as referring to any device that can be used to impart a radiation beam B with a pattern in its cross-section, such as to create a pattern in the target portion C of the substrate W. The pattern imparted to the radiation beam B can correspond to a particular functional layer in a device being created in the target portion C to form an integrated circuit.

[0030] The patterning device MA can be transmissive (as in lithographic apparatus **100'** of FIG. **1B**) or reflective (as in lithographic apparatus **100** of FIG. **1A**). Examples of patterning devices MA include reticles, masks, programmable mirror arrays, or programmable LCD panels. Masks are well known in lithography, and include mask types such as binary, alternating phase shift, or attenuated phase shift, as well as various hybrid mask types. An example of a programmable mirror array employs a matrix arrangement of small mirrors, each of which can be individually tilted so as to reflect an incoming radiation beam in different directions. The tilted mirrors impart a pattern in the radiation beam B, which is reflected by a matrix of small mirrors.

[0031] The term “projection system” PS can encompass any type of projection system, including refractive, reflective, catadioptric, magnetic, electromagnetic and electrostatic optical systems, or any combination thereof, as appropriate for the exposure radiation being used, or for other factors, such as the use of an immersion liquid on the substrate W or the use of a vacuum. A vacuum environment can be provided to the whole beam path with the aid of a vacuum wall and vacuum

pumps.

[0032] The lithographic apparatus can also be of a type wherein at least a portion of the substrate can be covered by a liquid having a relatively high refractive index, e.g., water, so as to fill a space between the projection system and the substrate. An immersion liquid can also be applied to other spaces in the lithographic apparatus, for example, between the mask and the projection system. Immersion techniques are well known in the art for increasing the numerical aperture of projection systems. The term “immersion” as used herein does not mean that a structure, such as a substrate, must be submerged in liquid, but rather only means that liquid is located between the projection system and the substrate during exposure.

[0033] Referring to FIGS. 1A and 1B, the illuminator IL receives a radiation beam from a radiation source SO. The source SO and the lithographic apparatus **100**, **100'** can be separate physical entities, for example, when the source SO is an excimer laser. In such cases, the source SO is not considered to form part of the lithographic apparatus **100** or **100'**, and the radiation beam B passes from the source SO to the illuminator IL with the aid of a beam delivery system BD (in FIG. 1B) including, for example, suitable directing mirrors and/or a beam expander. In other cases, the source SO can be an integral part of the lithographic apparatus **100**, **100'**, for example, when the source SO is a mercury lamp. The source SO and the illuminator IL, together with the beam delivery system BD, if required, can be referred to as a radiation system.

[0034] The illuminator IL can include an adjuster AD (in FIG. 1B) for adjusting the angular intensity distribution of the radiation beam. Generally, at least the outer and/or inner radial extent (commonly referred to as “ σ -outer” and “ σ -inner.” respectively) of the intensity distribution in a pupil plane of the illuminator can be adjusted. In addition, the illuminator IL can comprise various other components (in FIG. 1B), such as an integrator IN and a condenser CO. The illuminator IL can be used to condition the radiation beam B to have a desired uniformity and intensity distribution in its cross section.

[0035] Referring to FIG. 1A, the radiation beam B is incident on the patterning device (for example, mask) MA, which is held on the support structure (for example, mask table) MT, and is patterned by the patterning device MA. In lithographic apparatus **100**, the radiation beam B is reflected from the patterning device (for example, mask) MA. After being reflected from the patterning device (for example, mask) MA, the radiation beam B passes through the projection system PS, which focuses the radiation beam B onto a target portion C of the substrate W. With the aid of the second positioner PW and position sensor IF2 (for example, an interferometric device, linear encoder, or capacitive sensor), the substrate table WT can be moved accurately (for example, so as to position different target portions C in the path of the radiation beam B). Similarly, the first positioner PM and another position sensor IF1 can be used to accurately position the patterning device (for example, mask) MA with respect to the path of the radiation beam B. Patterning device (for example, mask) MA and substrate W can be aligned using mask alignment marks M1, M2 and substrate alignment marks P1, P2.

[0036] Referring to FIG. 1B, the radiation beam B is incident on the patterning device (for example, mask MA), which is held on the support structure (for example, mask table MT), and is patterned by the patterning device. Having traversed the mask MA, the radiation beam B passes through the projection system PS, which focuses the beam onto a target portion C of the substrate W.

[0037] The projection system PS projects an image of the mask pattern MP, where the image is formed by diffracted beams produced from the mark pattern MP by radiation from the intensity distribution, onto a photoresist layer coated on the substrate W. For example, the mask pattern MP can include an array of lines and spaces. A diffraction of radiation at the array and different from zeroth order diffraction generates diverted diffracted beams with a change of direction in a direction perpendicular to the lines. Undiffracted beams (i.e., so-called zeroth order diffracted beams) traverse the pattern without any change in propagation direction. The zeroth order

diffracted beams traverse an upper lens or upper lens group of the projection system PS, upstream of the pupil conjugate PPU of the projection system PS, to reach the pupil conjugate PPU. The portion of the intensity distribution in the plane of the pupil conjugate PPU and associated with the zeroth order diffracted beams is an image of the intensity distribution in the illumination system pupil IPU of the illumination system IL. The aperture device PD, for example, is disposed at or substantially at a plane that includes the pupil conjugate PPU of the projection system PS.

[0038] The projection system PS is arranged to capture, by means of a lens or lens group L, not only the zeroth order diffracted beams, but also first-order or first- and higher-order diffracted beams (not shown). In some embodiments, dipole illumination for imaging line patterns extending in a direction perpendicular to a line can be used to utilize the resolution enhancement effect of dipole illumination. For example, first-order diffracted beams interfere with corresponding zeroth-order diffracted beams at the level of the wafer W to create an image of the line pattern MP at highest possible resolution and process window (i.e., usable depth of focus in combination with tolerable exposure dose deviations). In some embodiments, astigmatism aberration can be reduced by providing radiation poles (not shown) in opposite quadrants of the illumination system pupil IPU.

[0039] With the aid of the second positioner PW and position sensor IF (for example, an interferometric device, linear encoder, or capacitive sensor), the substrate table WT can be moved accurately (for example, so as to position different target portions C in the path of the radiation beam B). Similarly, the first positioner PM and another position sensor (not shown in FIG. 1B) can be used to accurately position the mask MA with respect to the path of the radiation beam B (for example, after mechanical retrieval from a mask library or during a scan).

[0040] In general, movement of the mask table MT can be realized with the aid of a long-stroke module (coarse positioning) and a short-stroke module (fine positioning), which form part of the first positioner PM. Similarly, movement of the substrate table WT can be realized using a long-stroke module and a short-stroke module, which form part of the second positioner PW. In the case of a stepper (as opposed to a scanner), the mask table MT can be connected to a short-stroke actuator only or can be fixed. Mask MA and substrate W can be aligned using mask alignment marks M1, M2, and substrate alignment marks P1, P2. Although the substrate alignment marks (as illustrated) occupy dedicated target portions, they can be located in spaces between target portions (known as scribe-lane alignment marks). Similarly, in situations in which more than one die is provided on the mask MA, the mask alignment marks can be located between the dies.

[0041] Mask table MT and patterning device MA can be in a vacuum chamber V, where an in-vacuum robot IVR can be used to move patterning devices such as a mask in and out of vacuum chamber. Alternatively, when mask table MT and patterning device MA are outside of the vacuum chamber, an out-of-vacuum robot can be used for various transportation operations, similar to the in-vacuum robot IVR. Both the in-vacuum and out-of-vacuum robots need to be calibrated for a smooth transfer of any payload (e.g., mask) to a fixed kinematic mount of a transfer station.

[0042] The lithographic apparatus 100 and 100' can be used in at least one of the following modes:

[0043] 1. In step mode, the support structure (for example, mask table) MT and the substrate table WT are kept essentially stationary, while an entire pattern imparted to the radiation beam B is projected onto a target portion C at one time (i.e., a single static exposure). The substrate table WT is then shifted in the X and/or Y direction so that a different target portion C can be exposed. [0044] 2. In scan mode, the support structure (for example, mask table) MT and the substrate table WT are scanned synchronously while a pattern imparted to the radiation beam B is projected onto a target portion C (i.e., a single dynamic exposure). The velocity and direction of the substrate table WT relative to the support structure (for example, mask table) MT can be determined by the (de-) magnification and image reversal characteristics of the projection system PS. [0045] 3. In another mode, the support structure (for example, mask table) MT is kept substantially stationary holding a programmable patterning device, and the substrate table WT is moved or scanned while a pattern

imparted to the radiation beam B is projected onto a target portion C. A pulsed radiation source SO can be employed and the programmable patterning device is updated as required after each movement of the substrate table WT or in between successive radiation pulses during a scan. This mode of operation can be readily applied to maskless lithography that utilizes a programmable patterning device, such as a programmable mirror array.

[0046] Combinations and/or variations on the described modes of use or entirely different modes of use can also be employed.

[0047] In some embodiments, lithographic apparatus **100'** includes a deep ultraviolet (DUV) source, which is configured to generate a beam of DUV radiation for DUV lithography. A DUV source can be, for example, a gas discharge laser (e.g., an excimer laser).

Exemplary Lithographic Cell

[0048] FIG. 2 shows a lithographic cell **200**, also sometimes referred to a lithocell or cluster, according to some embodiments. Lithographic apparatus **100** or **100'** can form part of lithographic cell **200**. Lithographic cell **200** can also include one or more apparatuses to perform pre- and post-exposure processes on a substrate. Conventionally these include spin coaters SC to deposit resist layers, developers DE to develop exposed resist, chill plates CH, and bake plates BK. A substrate handler, or robot, RO picks up substrates from input/output ports I/O1, I/O2, moves them between the different process apparatuses and delivers them to the loading bay LB of the lithographic apparatus **100** or **100'**. These devices, which are often collectively referred to as the track, are under the control of a track control unit TCU, which is itself controlled by a supervisory control system SCS, which also controls the lithographic apparatus via lithography control unit LACU. Thus, the different apparatuses can be operated to maximize throughput and processing efficiency.

Exemplary Radiation Source

[0049] There exist many applications of pulsed-discharge radiation sources, for example, lithography, medical procedures, machining via laser ablation, laser imprinting, and more. A lithographic apparatus is one example in which a stable illumination source may be desirable. The illumination source can comprise precision optical assemblies that are sensitive to contaminants. FIG. 3 shows a radiation source **300**, according to some embodiments. In some embodiments, radiation source **300** is a pulsed-discharge radiation source, for example and without limitation, a gas discharge laser. Radiation source **300** comprises a gas chamber **302**, a window **304**, and a conduit system **306**. Radiation source **300** can further comprise one or more electrodes **310** (also “electrical connection”). Conduit system **306** can comprise a network of valves, conduits, and contaminant filters (not shown, but described in more detail in reference to FIG. 4).

[0050] In some embodiments, gas chamber **302** can confine a gas **308**. Gas **308** can comprise fluorine, neon, krypton, argon, and the like. Conduit system **306** is connected to gas chamber **302**. Conduit system **306** can allow management of gas **308** in gas chamber **302**. For example, conduit system **306** can direct a flow (e.g., circulation) of gas **308** to a filter within conduit system **306** to purify gas **308**. A voltage can be supplied to gas **308** (e.g., via one or more electrodes **310**) to generate radiation **312**. Window **304** can allow radiation **312** to exit gas chamber **302**.

[0051] FIG. 4 shows a radiation source **400**, according to some embodiments. In some embodiments, radiation source **400** shown in FIG. 4 can represent radiation source **300** shown in FIG. 3 in more detail. For example, FIG. 4 can show a more detailed view of conduit system **306**. Unless otherwise noted, elements of FIG. 4 that have similar reference numbers (e.g., reference numbers sharing the two right-most numeric digits) as elements of FIG. 3 can have similar structures and functions.

[0052] In some embodiments, radiation source **400** comprises a gas chamber **402**, a window **404**, and a conduit system (e.g., conduit system **306**, FIG. 3). Radiation source **400** can further comprise one or more electrodes **410**. Radiation source **400** can further comprise a window **418**. The structure and functions of window **418** can be similar to those of window **404**. The conduit system comprises a refill conduit **414** (also “refill path”) and a conduit **416**, and a contaminant filter **420**

(or simply filter). The conduit system can further comprise a unidirectional valve **422**, a conduit **424**, a unidirectional valve **426**, and any combinations thereof.

[0053] In some embodiments, gas chamber **402** can confine a gas **408**. The conduit system can be connected to gas chamber **402** to allow circulation of gas **408**, for example, during operation to generate radiation **412**. For example, the conduit system can circulate gas **408** to contaminant filter **420** that is connected to gas chamber **402** (gas flow direction designated by arrow **428**).

Contaminant filter **420** can remove contaminant particles **432** from gas chamber **402**. Conduit **416** connects contaminant filter **420** back to gas chamber **402** and the gas flow is such that clean, filtered gas can blow on window **404** (gas flow direction designated by arrow **430**). Window **404** can confine gas **408** from an environment external to gas chamber **402**. A pressure differential device (not shown) can be used to cause gas flow in radiation source **400**. For example, a blower can be inside gas chamber **402**. Contaminant filter **420** may intercept a portion of the flow circulating through gas chamber **402** and redirect the flow toward windows **404** and **418** after removing contaminant particles **432**. Gas **408** can be supplied or evacuated using refill conduit **414** that is connected to gas chamber **402**. That is, refill conduit **414** can allow replacement of gas **408**. It should be appreciated that the plumbing configuration shown in FIG. **4** is provided as a non-limiting example. For example, plumbing configurations can be envisaged that use more or fewer conduits. T-junctions, valves, and the like to achieve cleanliness of sensitive optical components, such as window **404** and window **418**.

[0054] In some embodiments, arrows **428**, **430**, and **434** represent gas flows during operation of radiation source **400**, for example, when generating radiation **412**. To generate radiation **412**, a voltage can be supplied to gas **408**, for example, via one or more electrodes **410**. Radiation **412** can have properties that depend on the applied voltage (e.g., an electrical pulse for a pulse of radiation). Windows **404** and **418** can allow radiation **412** to travel between gas chamber **402** and the environment external to gas chamber **402**. In some embodiments, radiation source **400** is a gas discharge laser. Radiation source **400** can comprise an optical reflector **436** and a partial optical reflector **438**. Optical reflector **436** and partial optical reflector **438** together function as an optical resonator. The optical resonator, in combination with a gain medium (e.g., gas **408**) allow for amplification of radiation **412** as it travels back and forth between optical reflector **436** and partial optical reflector **438**. Radiation source **400** can then output a beam of radiation **440** via transmission at partial optical reflector **438**.

[0055] In some embodiments, the expected lifetime of gas chamber **402** depends on the first critical component to become inoperable via wear and tear. One example of a component subject to wear and tear is one or more electrodes **410**. During operation of radiation source **400**, one or more electrodes **410** interact with gas **408**. The interaction causes the electrode material to combine with gas **408** and detach from one or more electrodes **410**, effectively eroding one or more electrodes **410**. Such erosion is expected and has a predictable erosion rate. The more radiation source **400** is operated, the more the electrodes erode. A benchmark for the lifetime of at least one or more electrodes **410** can be prescribed as the time it takes for one or more electrodes to go from new to eroded beyond the point of operability (e.g., can be measured in number of pulses generated throughout the lifetime). An undesirable behavior of radiation source **400** is one where a component fails unpredictably before the prescribed lifetime.

[0056] In some embodiments, one or more of contaminant particles **432** have a probability of settling on window **404** due to unintended gas flows. For example, it was mentioned earlier that, arrows **428**, **430**, and **434** represent gas flows during operation of radiation source **400**. As radiation source **400** is operated, the quality of gas **408** degrades (e.g., becomes spent). Therefore, gas **408** can be replaced with fresh new gas (also “refill gas”) by accessing gas chamber **402** using refill conduit **414**. The refill gas may be of the same type as gas **408** in its unspent state or comprised of a different unspent gas. In some embodiments, a procedure to replace gas **408** comprises evacuating gas **408** using refill conduit **414** and then inserting the refill gas again using refill conduit **414**. The

procedure to replace gas **408** can stir up contaminant particles **432** that would normally be settled at the bottom of gas chamber **402**. The stirred up contaminant particles **432** can have a higher probability of landing on windows **404** and **418**. Moreover, the evacuation of gas **408** can pull contaminant particles **432** into refill conduit **414** and then subsequent insertion of the refill gas via the same conduit can blow contaminant particles **432** all around gas chamber **402**, some even landing on windows **404** and **418**.

[0057] In some embodiments, while window **404** is transparent and allows radiation **412** to pass through, contaminant particles can absorb a considerable amount of energy from radiation **412**, thereby heating any contaminant particles **432** that have settled on window **404** and transferring that heat to window **404**. The energy density in radiation **412** can be high enough to damage window **404** via heating of contaminant particles **432** on window **404**. The failure of window **404** can occur unpredictably and well before exhausting the lifetime of one or more electrodes **410**. In high-volume production of ICs, unpredictable machine downtime (e.g., unscheduled maintenance) is highly detrimental due to unexpected loss of production time. In a scenario where failure is predictable (e.g., occurring at the prescribed lifetime, scheduled maintenance), backup parts and procedures can be prepared. Structures and functions described in embodiments herein can reduce the probability of unpredictable early failure of pulsed-discharge radiation sources, and thus improve the average lifetime and reliability of pulsed-discharge radiation sources.

[0058] In some embodiments, the conduit system is configured to direct a flow of one of the refill gas, gas **408**, or both the refill gas and gas **408**. The direction of gas flow can change depending on the state of operation of radiation source **400** (e.g., during operation or during a refill procedure) based on the plumbing configuration of the conduit system. The gas flow can be manipulated in such a way so as to avoid stirring up or otherwise directing contaminant particles **432** toward windows **404** and **418**. By manipulating the gas flows in this manner, the conduit system can increase the usable lifetime of windows **404** and **418**. The useable lifetime of gas chamber **402**, radiation source **400**, and the like, are also improved since unexpected and costly disassembly can be avoided (since it may be more efficient to just outright replace a radiation source rather than disassemble it, the usable lifetime of a radiation source can be dictated by failure of just a window).

[0059] In some embodiments, unidirectional valve **422** can be disposed to intersect conduit **416**. Unidirectional valve **422** can comprise a check valve, for example, a ball check valve, a flap check valve, a spring check valve, a gravity check valve, and the like. A check valve is a valve that closes to prevent a backward flow. Unidirectional valve **422** can represent a system of check valves that combines structures and/or features of any of the check valves mentioned above. For example, a ball check valve may be used vertically, whereby the ball is pushed down and closed by gravity (e.g., gravity check valve). While a check valve uses the pressure of gas travelling in the “wrong” direction to shut itself, the extra force (e.g., from gravity) on the shutting mechanism can provide a crack pressure threshold in order for a gas to push the check valve open. In some embodiments, unidirectional valve **422** can be a user-adjustable valve (e.g., an electrically actuated valve, a ball valve with an adjustable angle with respect to the direction of gravity, a pneumatic valve, and the like).

[0060] In some embodiments, during an evacuation of gas **408**, unidirectional valve **422** can prevent gas flow (represented by arrow **430**) from reversing. For example, gas **408** from gas chamber **402** is prevented from travelling toward window **404**. In this manner, the probability of contaminant particles **432** contacting window **404** is reduced.

[0061] It was mentioned earlier that refill conduit **414** can become contaminated with contaminant particles **432**. In order to avoid spreading contaminant particles **432**, in some embodiments, the conduit system can comprise a separate evacuation conduit **442** (also “evacuation path”) dedicated for evacuation of gas **408**. In this scenario, refill conduit **414** can be dedicated for insertion of a refill gas. Since the evacuation and refill of gas are handled with separate conduits, this configuration is able to avoid blowing the contaminants in evacuation conduit **442** back into gas

chamber **402**. It should be appreciated that the location of refill conduit **414** and evacuation conduit **442** is not limited to the representation in FIG. **4** and that their locations can be chosen so as to minimize the likelihood of spreading contaminant particles **432** around gas chamber **402** (e.g., the locations of refill conduit **414** and evacuation conduit **442** can be interchanged).

[0062] In some embodiments, additional elements may be comprised by the conduit system that allow for manipulation of the flow of gas **408** and/or the refill gas. The structures of the additional elements are shown in FIG. **4** while the functions are described in more detail in reference to FIG. **5**. The conduit system can further comprise conduit **444** (also “bypass conduit”), unidirectional valve **446**, conduit **448** (also “bypass conduit”), and unidirectional valve **450**. Unidirectional valve **452** can be disposed to intersect refill conduit **414**. Unidirectional valve **452** can prevent gas from entering gas chamber **402** via refill conduit **414**. Any combinations of the structures mentioned above can be employed to achieve a desired flow direction of gas **408** and the refill gas.

[0063] It should be appreciated that, in some embodiments, conduit **424**, conduit **448**, unidirectional valve **426**, unidirectional valve **450**, and window **418** may be structured and configured similar to conduit **416**, conduit **444**, unidirectional valve **422**, unidirectional valve **446**, and window **404**, respectively. For example, the similarities may be a structural and/or functional symmetry, exactly or approximately.

[0064] FIGS. **5A**, **5B**, and **5C** show a portion of a radiation source **500** according to some embodiments. In some embodiments, radiation source **500** can also represent radiation source **300** (FIG. **3**) and/or radiation source **400** (FIG. **4**) in more detail. For example, FIG. **5** can show a more detailed view of conduit system **306** and its functions. Unless otherwise noted, elements of FIG. **5** that have similar reference numbers (e.g., reference numbers sharing the two right-most numeric digits) as elements of FIGS. **3** and **4** can have similar structures and functions.

[0065] Referring to FIG. **5A**, flows of a gas **508** are shown (indicated by arrows and also shown as gas particles and not to be confused with contaminant particles **432** of FIG. **4**). Radiation source **500** comprises a gas chamber **502**, a window **504**, and a conduit system (e.g., conduit system **306**, FIG. **3**). Radiation source **500** can further comprise a window **518**. The conduit system can comprise a contaminant filter **520**, a refill conduit **514**, a conduit **516**, a conduit **524**, a unidirectional valve **522**, and a unidirectional valve **526**. The conduit system can further comprise, a conduit **544**, a conduit **548**, a unidirectional valve **546**, a unidirectional valve **550**, and a unidirectional valve **552**. Some of these elements have already been described in reference to their respective counterparts in FIG. **4** (e.g., referenced by numbers sharing the two right-most numeric digits).

[0066] In some embodiments, a flow of gas **508** (represented by arrows) is as shown in FIG. **5A** during operation of radiation source **500** (e.g., generating radiation). Contaminant filter **520** can remove contaminants (e.g., contaminant particles **432** (FIG. **4**)) from gas chamber **502**. Conduit **516** connects contaminant filter **520** back to gas chamber **502** and the gas flow is such that clean, filtered gas can blow on window **504**. To facilitate the direction of flow, unidirectional valve **522** is shown in the open state to allow filtered gas to blow on window **504**. Conversely, unidirectional valve **522** can close to prevent gas flow from reversing to prevent drawing contaminants from gas chamber **502** toward window **504**. Though functions are described in reference to contaminants on window **504**, it is to be appreciated that similar or symmetric processes are employed with respect to window **518** using corresponding conduits and valves.

[0067] In reference to FIG. **5B**, in some embodiments, a flow of gas **508** during evacuation of gas **508** is represented by the arrows shown. In instances where the gas flow is shown to move away in both directions from a unidirectional valve (e.g., unidirectional valves **546**, and **550**), such valves may be closed when the gas flow is in the direction against the directionality of the unidirectional valve or the pressure of the gas flow is insufficient to overcome the cracking-pressure threshold of the unidirectional valve. To facilitate evacuation of gas **508**, unidirectional valve **552** is shown in the open state to allow gas **508** to exit gas chamber **502** in a manner that prevents gas **508** and any

contaminants from flowing toward window **504**.

[0068] In reference to FIG. **5C**, in some embodiments, a flow of refill gas during refilling gas chamber **502** is represented by the arrows shown. To facilitate refilling gas chamber **502**, unidirectional valve **546** is shown in the open state to allow the refill gas to enter gas chamber **502**. Unidirectional valve **552** is in the closed state so as to prevent a flow of gas from gas chamber **502** toward window **504** (preventing contaminants present in gas chamber **502** from flowing toward window **504**). In this configuration, the refill gas (which is pure and contaminant-free) enters gas chamber **502** via conduit **544**. Conduit **544** can be a bypass conduit that bypasses the orifice that connects gas chamber **502** and refill conduit **514**. That is, conduit **544** makes direct connection between conduit **516** and refill conduit **514**. The uncontaminated refill gas (or filtered gas **508** in FIG. **5A**) can flow across windows **504** and **518** while not depositing any dust on windows **504** and **518**. The uncontaminated refill gas can also exert a pressure on window **504** that can blow away any contaminants present on window **504**. In this manner, the usable lifetime of at least window **504** can be increased owing to the reduced probability of contaminants on window **504** absorbing radiation energy.

[0069] FIG. **6** shows a unidirectional valve **600**, according to some embodiments. Unidirectional valve **600** comprises a conduit section **602** and a flap **604**. Unidirectional valve **600** can further comprise a hinge **606**. Hinge **606** can attach flap **604** to conduit section **602**. If hinge **606** is omitted, flap **604** can be a flexible flap (e.g., flexure) and attach directly to conduit section **602**. Hinge **606** can be spring loaded such that a cracking-pressure threshold can be defined (e.g., valve opens only if flow pressure exceeds a predetermined amount). A flexure flap can also define a cracking-pressure threshold. Arrow **608** represents a direction of flow that is allowed by unidirectional valve **600**.

[0070] FIG. **7** shows a cross-section of a unidirectional valve **700**, according to some embodiments. Unidirectional valve **700** comprises a conduit section **702** and a ball **704**. Ball **704** can be spring loaded such that a cracking-pressure threshold can be defined. Unidirectional valve **700** can also define a cracking-pressure threshold even without a spring (e.g., using gravity and the weight of ball **704**). Arrow **708** represents a direction of flow that is allowed by unidirectional valve **700**.

[0071] FIG. **8** shows method steps for performing functions described herein, according to some embodiments. The method steps of FIG. **8** can be performed in any conceivable order and it is not required that all steps be performed. Moreover, the method steps of FIG. **8** described below merely reflect an example of steps and are not limiting. That is, further method steps and functions may be envisaged based upon embodiments described in reference to FIGS. **1-7**.

[0072] At step **802**, radiation is generated using a pulsed-discharge radiation system.

[0073] At step **804**, a gas and contaminants produced during the generating are confined using a gas chamber.

[0074] At step **806**, the gas is isolated from an environment external to the gas chamber using a window.

[0075] At step **808**, the radiation is allowed to travel between the gas chamber and the environment using the window.

[0076] At step **810**, the gas is replaced using a refill path.

[0077] At step **812**, a flow of one of a refill gas, the gas, or the refill gas and the gas during a refill operation is directed to prevent the contaminant from contacting the window.

[0078] Although specific reference can be made in this text to the use of lithographic apparatus in the manufacture of ICs, it should be understood that the lithographic apparatus described herein may have other applications, such as the manufacture of integrated optical systems, guidance and detection patterns for magnetic domain memories, flat-panel displays, LCDs, thin-film magnetic heads, etc. The skilled artisan will appreciate that, in the context of such alternative applications, any use of the terms “wafer” or “die” herein can be considered as synonymous with the more general terms “substrate” or “target portion”, respectively. The substrate referred to herein can be

processed, before or after exposure, in for example a track unit (a tool that typically applies a layer of resist to a substrate and develops the exposed resist), a metrology unit and/or an inspection unit. Where applicable, the disclosure herein can be applied to such and other substrate processing tools. Further, the substrate can be processed more than once, for example in order to create a multi-layer IC, so that the term substrate used herein may also refer to a substrate that already contains multiple processed layers.

[0079] It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present disclosure is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0080] The terms “radiation,” “beam,” “light,” “illumination,” and the like as used herein may encompass all types of electromagnetic radiation, for example, ultraviolet (UV) radiation (for example, having a wavelength λ of 365, 248, 193, or 157 nm). DUV generally refers to radiation having wavelengths ranging from 130 nm to 428 nm, and in some embodiments, an excimer laser can generate DUV radiation used within a lithographic apparatus. It should be appreciated that radiation having a wavelength in the range of, for example, 130-428 nm relates to radiation with a certain wavelength band, of which at least part is in the range of 130-428 nm.

[0081] The term “substrate” as used herein describes a material onto which material layers are added. In some embodiments, the substrate itself can be patterned and materials added on top of it may also be patterned, or may remain without patterning.

[0082] Although specific reference can be made in this text to the use of the apparatus and/or system according to the present disclosure in the manufacture of ICs, it should be explicitly understood that such an apparatus and/or system has many other possible applications. For example, it can be employed in the manufacture of integrated optical systems, guidance and detection patterns for magnetic domain memories, LCD panels, thin-film magnetic heads, etc. The skilled artisan will appreciate that, in the context of such alternative applications, any use of the terms “reticle,” “wafer,” or “die” in this text should be considered as being replaced by the more general terms “mask,” “substrate,” and “target portion,” respectively.

[0083] While specific embodiments of the disclosure have been described above, it will be appreciated that embodiments of the present disclosure may be practiced otherwise than as described. The descriptions are intended to be illustrative, not limiting. Thus it will be apparent to one skilled in the art that modifications may be made to the disclosure as described without departing from the scope of the claims set out below.

[0084] It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

[0085] The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

[0086] The foregoing description of the specific embodiments will so fully reveal the general nature of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein.

[0087] Other aspects of the invention are set out in the following numbered clauses. [0088] 1. A pulsed discharge radiation source configured to generate radiation, the pulsed-discharge radiation

system comprising: [0089] a gas chamber configured to confine a gas and a contaminant produced during generation of the radiation; [0090] a window configured to isolate the gas from an environment external to the gas chamber and to allow the radiation to travel between the gas chamber and the environment; and [0091] a conduit system comprising: [0092] a refill path configured to allow a replacement of the gas; and [0093] a conduit configured to circulate the gas to or from the gas chamber during the generation of the radiation, [0094] wherein the conduit system is configured to direct a flow of one of a refill gas, the gas, or the refill gas and the gas at least during a refill operation to prevent the contaminant from contacting the window, whereby the conduit system increases usable lifetime of at least the window. [0095] 2. The pulsed-discharge radiation source of clause 1, wherein the conduit system further comprises a valve disposed to intersect the conduit and configured to prevent a flow of at least a portion of the contaminant toward the window. [0096] 3. The pulsed-discharge radiation source of clause 2, wherein the valve comprises a unidirectional valve. [0097] 4. The pulsed-discharge radiation source of clause 3, wherein the unidirectional valve comprises at least one of a ball check valve, flap check valve, spring check valve, and gravity check valve. [0098] 5. The pulsed-discharge radiation source of clause 2, wherein the valve comprises a user-adjustable valve. [0099] 6. The pulsed-discharge radiation source of clause 1, wherein: [0100] the conduit system further comprises a bypass conduit and a unidirectional valve disposed to intersect the bypass conduit; [0101] the bypass conduit connects the conduit to the refill path; and [0102] the unidirectional valve is configured to prevent a flow of the gas from the gas chamber and toward the window. [0103] 7. The pulsed-discharge radiation source of clause 1, wherein the refill path is further configured to allow evacuation of the gas. [0104] 8. The pulsed-discharge radiation source of clause 1, wherein the conduit system further comprises an evacuation path configured to allow evacuation of the gas to prevent the contaminant from entering the refill path. [0105] 9. The pulsed-discharge radiation source of clause 1, further comprising an electrical connection configured to deliver an electrical pulse to the gas to generate the radiation. [0106] 10. The pulsed-discharge radiation source of clause 1, wherein the radiation comprises DUV radiation. [0107] 11. A method comprising: [0108] generating radiation using a pulsed-discharge radiation system; [0109] confining a gas and contaminants produced during the generating, using a gas chamber; [0110] isolating the gas from an environment external to the gas chamber using a window; [0111] allowing the radiation to travel between the gas chamber and the environment using the window; [0112] replacing the gas using a refill path; [0113] circulating the gas to or from the gas chamber during the generating; and [0114] directing a flow of one of a refill gas, the gas, or the refill gas and the gas during a refill operation to prevent the contaminants from contacting the window. [0115] 12. The method of clause 11, further comprising preventing a flow of at least a portion of the contaminants toward the window using a unidirectional valve. [0116] 13. The method of clause 12, further comprising preventing a flow of the gas from the gas chamber toward the window using a further unidirectional valve, wherein the further unidirectional valve is disposed to intersect a bypass conduit that connects the conduit to the refill path. [0117] 14. The method of clause 11, further comprising evacuating the gas using the refill path. [0118] 15. The method of clause 11, further comprising evacuating the gas using an evacuation path. [0119] 16. The method of clause 11, wherein the generating comprises delivering an electrical pulse to the gas using an electrical connection. [0120] 17. The method of clause 11, wherein the radiation comprises DUV radiation. [0121] 18. A lithographic apparatus comprising: [0122] a pulsed-discharge illumination system configured to generate radiation to illuminate a pattern of a patterning device, the illumination system comprising: [0123] a gas chamber configured to confine a gas and contaminants produced during generation of the radiation; [0124] a window configured to isolate the gas from an environment external to the gas chamber and to allow the radiation to travel between the gas chamber and the environment; and [0125] a conduit system comprising: [0126] a refill path configured to allow a replacement of the gas; and [0127] a conduit configured to circulate the gas to or from the gas chamber during the generation of the radiation,

[0128] wherein the conduit system is configured to direct a flow of one of a refill gas, the gas, or the refill gas and the gas at least during a refill operation to prevent the contaminant from contacting the window, [0129] whereby the conduit system increases usable lifetime of at least the window; and a projection system configured to project an image of the pattern onto a substrate. [0130] 19. The lithographic apparatus of clause 18, wherein the conduit system further comprises a unidirectional valve disposed to intersect the conduit and configured to prevent a flow of at least a portion of the contaminant toward the window. [0131] 20. The lithographic apparatus of clause 19, wherein the unidirectional valve comprises at least one of a ball check valve, flap check valve, spring check valve, and gravity check valve. [0132] 21. The lithographic apparatus of clause 18, wherein: [0133] the conduit system further comprises a bypass conduit and a unidirectional valve disposed to intersect the bypass conduit; [0134] the bypass conduit connects the conduit to the refill path; and [0135] the unidirectional valve is configured to prevent a flow of the gas from the gas chamber and toward the window. [0136] 22. The lithographic apparatus of clause 18, wherein the conduit system further comprises an evacuation path configured to allow evacuation of the gas to prevent the contaminant from entering the refill path. [0137] 23. The lithographic apparatus of clause 18, wherein the radiation comprises DUV radiation. [0138] The breadth and scope of the protected subject matter should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A pulsed-discharge radiation source configured to generate radiation, the pulsed-discharge radiation source comprising: a chamber configured to confine a gas and to contain contaminants produced during generation of the radiation; a window configured to isolate the gas from an environment external to the chamber and to allow the radiation to travel between the chamber and the environment; and a recirculation conduit system configured to recirculate the gas from and back to the chamber, the recirculation conduit system comprising a contaminant filter arranged to receive the gas and the contaminants from the chamber and a conduit arranged to convey the gas from the contaminant filter to the chamber at the window, the conduit including a unidirectional valve arranged to prevent gas backward flow from the chamber at the window to the contaminant filter.
2. The pulsed-discharge radiation source of claim 1, wherein the unidirectional valve comprises at least one of a ball check valve, flap check valve, spring check valve, and gravity check valve.
3. The pulsed-discharge radiation source of claim 2, wherein the unidirectional valve comprises a user-adjustable valve.
4. The pulsed-discharge radiation source of claim 1, further comprising a refill conduit system configured to allow replacement of the gas in the chamber.
5. The pulsed-discharge radiation source of claim 4, wherein the refill conduit system further comprises an evacuation path configured to allow evacuation of the gas to prevent the contaminants from entering the refill path.
6. The pulsed-discharge radiation source of claim 1, further comprising an electrical connection configured to deliver an electrical pulse to the gas to generate the radiation.
7. The pulsed-discharge radiation source of claim 1, wherein the radiation comprises deep ultraviolet radiation.
8. A method comprising: generating radiation using a pulsed-discharge radiation system; confining a gas and contaminants, the contaminants being produced during the generating, using a chamber; isolating the gas from an environment external to the chamber using a window; allowing the radiation to travel between the gas chamber and the environment using the window; recirculating the gas in the chamber by causing the gas to flow on a path from the chamber through a

contaminant filter and back to the chamber at a position near the window, while preventing a backward flow of the gas using a unidirectional valve.

9. The method of claim 8, further comprising evacuating the gas using an evacuation path.

10. The method of claim 8, wherein the generating comprises delivering an electrical pulse to the gas using an electrical connection.

11. The method of claim 8, wherein the radiation comprises deep ultraviolet radiation.

12. A lithographic apparatus comprising: a pulsed-discharge illumination system configured to generate radiation to illuminate a pattern of a patterning device, the illumination system comprising a chamber configured to confine a gas and to confine contaminants produced during generation of the radiation, a window configured to isolate the gas from an environment external to the chamber and to allow the radiation to travel between the chamber and the environment, and a recirculation conduit system configured to recirculate the gas from and then back to the chamber, the recirculation conduit system comprising a contaminant filter arranged to receive the gas and the contaminants from the chamber and a conduit arranged to convey filtered gas from the contaminant filter to the chamber at the window, the conduit including a unidirectional valve arranged to prevent gas backward flow from the chamber at the window to the contaminant filter; and a projection system configured to project an image of the pattern onto a substrate.

13. The lithographic apparatus of claim 12, wherein the unidirectional valve comprises at least one of a ball check valve, flap check valve, spring check valve, and gravity check valve.

14. The lithographic apparatus of claim 12, wherein the pulsed-discharge illumination system further comprises an evacuation path configured to allow evacuation of the gas to prevent the contaminants from entering the refill path.

15. The lithographic apparatus of claim 12, wherein the radiation comprises deep ultraviolet radiation.

16. A pulsed-discharge radiation source configured to generate deep ultraviolet radiation, the pulsed-discharge radiation source comprising: a chamber configured to confine a gas and to confine contaminants produced during generation of the radiation; a window arranged at a window position in the chamber and configured to isolate the gas from an environment external to the chamber and to allow the radiation to travel between the chamber and the environment; and a recirculation conduit system configured to recirculate the gas from the chamber back to the chamber, the recirculation conduit system comprising a contaminant filter arranged to receive the gas and the contaminants from the chamber and a conduit arranged to convey filtered gas from the contaminant filter to the window position, the conduit including a unidirectional valve arranged to prevent gas backward flow from the window position to the contaminant filter.

17. A pulsed-discharge radiation source configured to generate deep ultraviolet radiation, the pulsed-discharge radiation source comprising: a chamber configured to confine a gas and to confine contaminants produced during generation of the radiation; a window arranged at a window position in the chamber and configured to isolate the gas from an environment external to the chamber and to allow the radiation to travel between the chamber and the environment; and a recirculation conduit system configured to recirculate the gas from the chamber back to the chamber at the window position, the recirculation conduit system comprising a conduit arranged to convey gas to the window position, the conduit including a unidirectional valve arranged to prevent gas backward flow from the window position to the conduit.
